Constellations of Prime Chains on a Logarithmic Spiral: Geometric Symmetries of Cunningham and Collatz-type Chains

March 8, 2026 — Hiroshi Harada

1. Overview

This project visualizes the geometric symmetries hidden within seemingly chaotic "prime chains."

In a polar coordinate system using natural logarithms, each number is plotted with radius $\boldsymbol{r = \ln(n)}$ and angle $\boldsymbol{\theta = 2\pi \ln(n)}$. In this logarithmic space, the growth rate of the recurrence relation directly determines the shape of the figure. When a recurrence multiplies n by approximately k, the angular displacement becomes:

$$\Delta\theta \approx 2\pi \ln(k)$$

Because $\ln(k)$ happens to be extremely close to a rational number p/q, the sequence completes almost p rotations in q steps. As a result, self-similar expanding figures ("constellations") resembling triangles or pentagrams naturally emerge.

2. The Cunningham Triangle

– Recurrence Formula: $\boldsymbol{n \to 2n \pm 1}$

– Geometric Observation: The growth rate is approximately 2. Since $\ln(2) \approx 0.693$ is close to the rational number $2/3$, the sequence rotates about two full turns (4π) every three steps, forming a self-similar expanding "triangle."

– Included Data: Two of the world's longest known Cunningham chains

Length-17 (1st kind): $\boldsymbol{n_0 = 2759832934171386593519}$

Length-19 (2nd kind): $\boldsymbol{n_0 = 7991019772166787018701610 1}$

3. The Collatz Pentagram

– Recurrence Formula: $\boldsymbol{n \to (3n+1)/2}$, applied only when the result is prime

– Geometric Observation: The growth rate is approximately 1.5. Since $\ln(1.5) \approx 0.405$ is close to the rational number $2/5$, the sequence rotates about two full turns (4π) every five steps, producing a beautifully expanding "pentagram" whose trajectory intersects itself.

– Included Data: Three Collatz-type prime chains, including newly discovered ones

Length-7: $\boldsymbol{n_0 = 89599}$

Length-8: $\boldsymbol{n_0 = 44102911}$ **(New)**

Length-9: $\boldsymbol{n_0 = 35014031359}$ **(New)**

4. Results

Using the ultra-fast multiple-precision arithmetic library gmpy2, a full search was performed up to 10^{12}. No Collatz-type prime chains of length-10 (L10) or greater were found below this bound. This result provides an important lower bound for future investigations.

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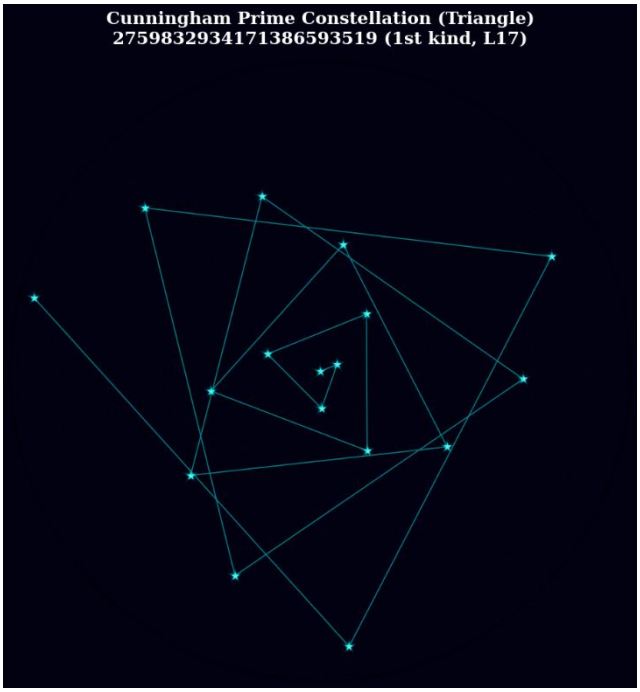
Code and data are available with the Zenodo submission.



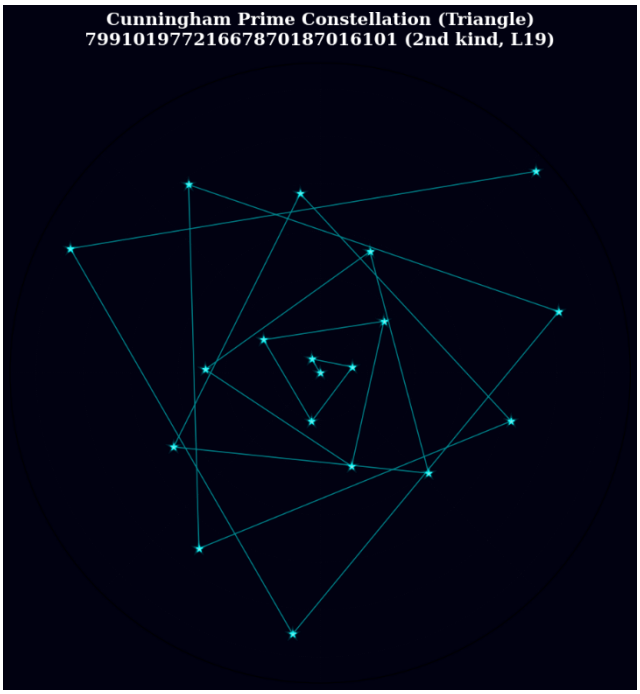


Figure 1. The Cunningham Triangle.

A logarithmic-spiral plot of a Cunningham prime chain. Because $ln(2) \approx 2/3$, the sequence rotates nearly two full turns every three steps, producing a self-similar expanding triangular pattern.

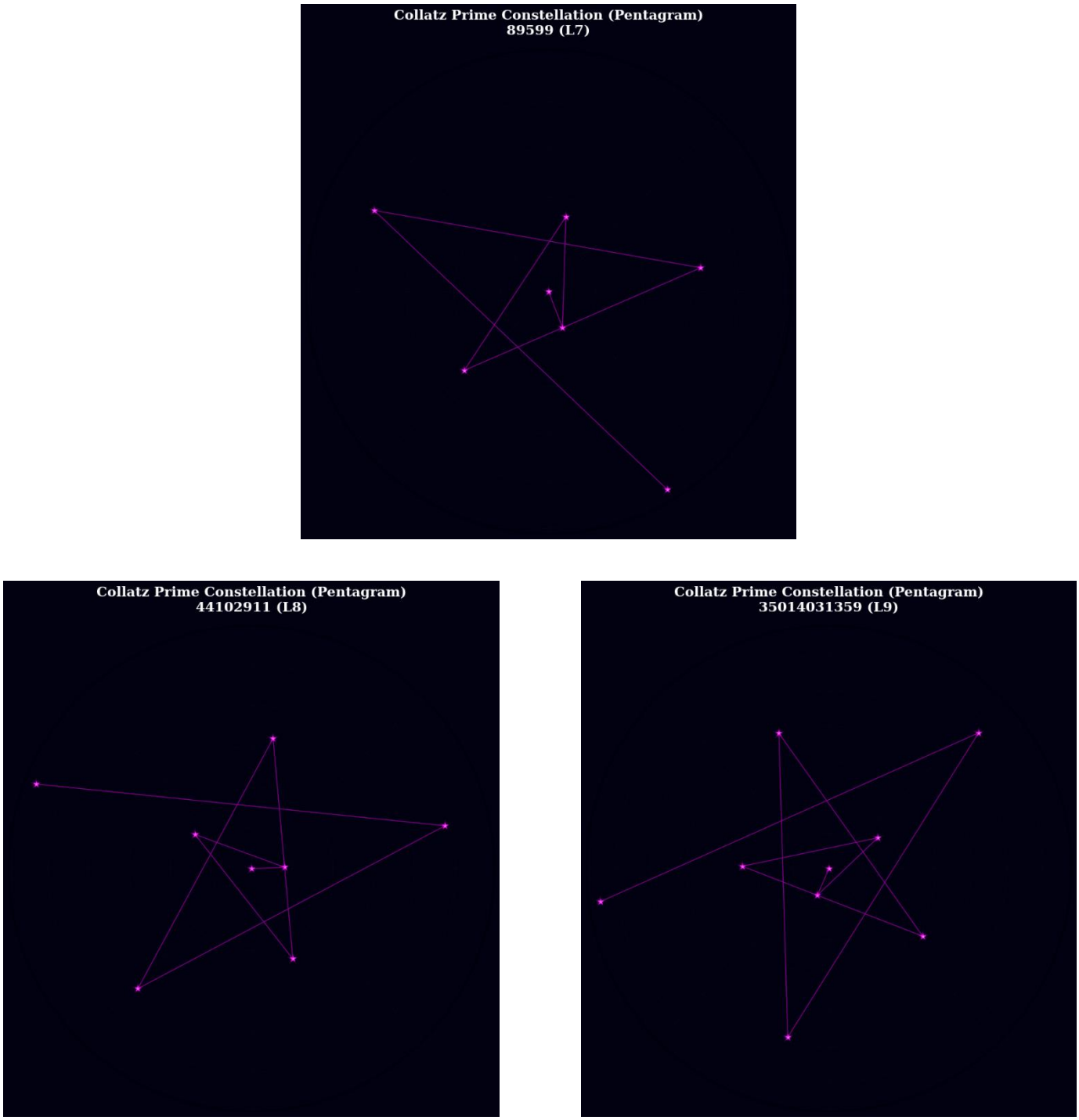


Figure 2. The Collatz Pentagram.

A logarithmic-spiral plot of a Collatz-type prime chain. Since $ln(1.5) \approx 2/5$, the sequence rotates about two full turns every five steps, forming an intersecting, self-similar pentagram.